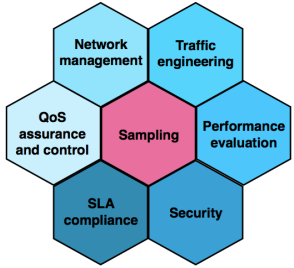


Enhancing Traffic Sampling Scope and Efficiency

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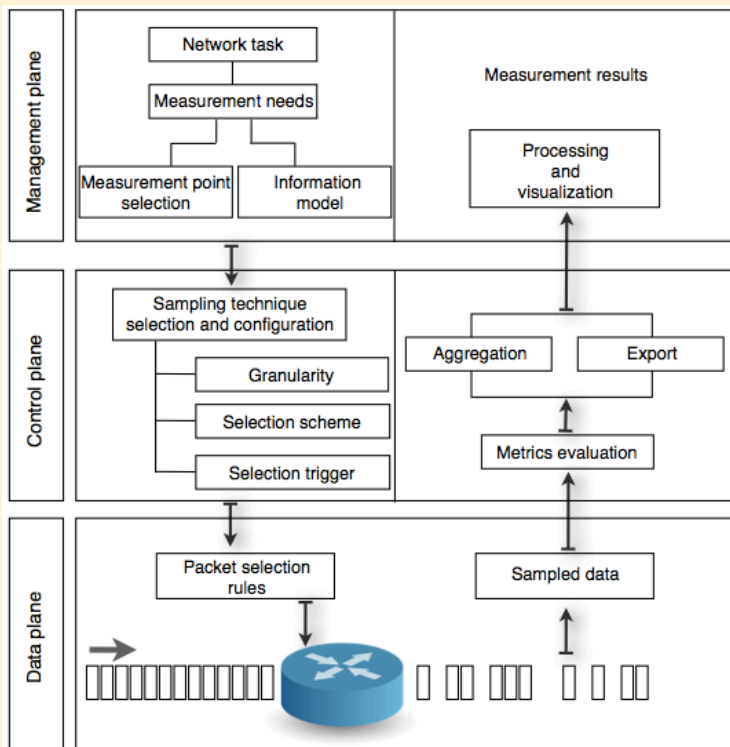
MOTIVATION:

- Traffic Sampling is a crucial step towards scalable network measurements, enclosing manifold challenges.
- The wide variety of foreseeable sampling scenarios demands for a modular view of sampling components.
- Articulating the measurement scope, the required information model and the adequate sampling strategy is a major design issue for achieving an encompassing and efficient sampling solution.

WE PROPOSE:

- A layered sampling-based *measurement architecture*.
- A *taxonomy* of classical and new sampling techniques distinguishing their inner features.
- A *framework* allowing a flexible combination of sampling features.
- A new *multiadaptive sampling technique* able to improve the trade-off between accuracy and overhead.

SAMPLING-BASED MEASUREMENT ARCHITECTURE



DATA PLANE

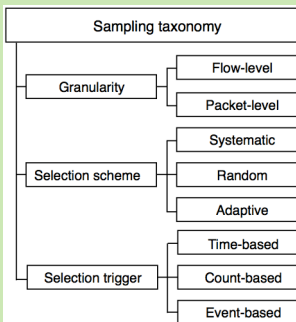
- The traffic is selected from network interfaces by applying the sample rules defined in the control plane.
- The packet fields are collected following the data model definition.
- The unprocessed sampled traffic is reported to the control plane to be analyzed, simplifying the data plane.

MANAGEMENT PLANE

- *Measurement needs* are identified based on specific requirements of each network task, then one or more measurement points are selected.
- An *information model*, able to define managed objects independently of specific implementation or protocol, is identified.
- Measurement results reported by the control plane are processed and visualized.

CONTROL PLANE

- A *modular design* allows a flexible sampling technique selection and configuration.
- The *proposed taxonomy* supports the configuration of sampling techniques adjusted to each traffic/service measurement scenario.

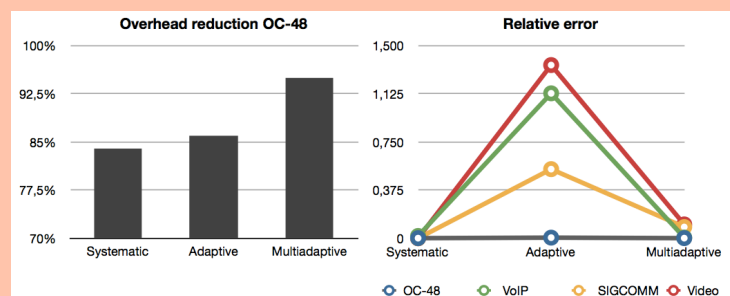


- **Granularity** identifies the atomicity of the element under analysis.
- **Selection scheme** corresponds to the function that defines which packets will be selected and collected.
- **Selection trigger** is used to decide the spatial and temporal sample boundaries.

- The sampled data received from the data plane is processed and relevant metrics are estimated according to the network task measurement needs.
- These metrics are then aggregated (both in time and space) and exported following IETF guidelines (RFC6728, RFC6313), using e.g. IPFIX specifications.

ONGOING DEVELOPMENTS

- A *framework* has been developed based on the proposed taxonomy, allowing a flexible combination of the sampling features defined above. It may be applied to both online and offline measurement scenarios.
- A new *Multiadaptive sampling technique* is proposed and compared to conventional sampling techniques using real traffic traces (OC48 from a US ISP, SIGCOMM08, SIP VoIP and Video streaming).
- As illustrated this technique brings clearly an added value regarding the reduction of sampling data without compromising the relative error in throughput estimation.



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